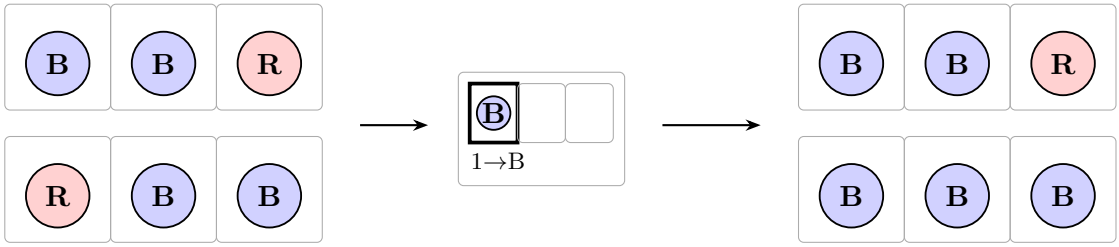
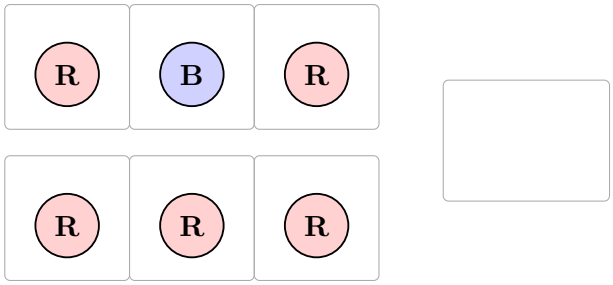
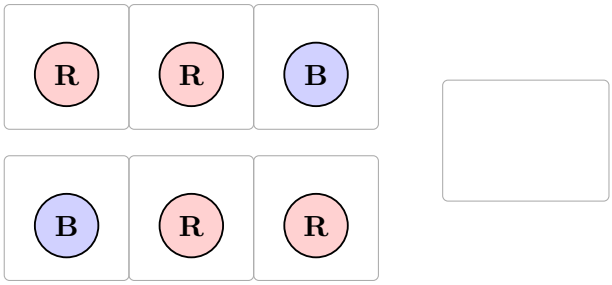
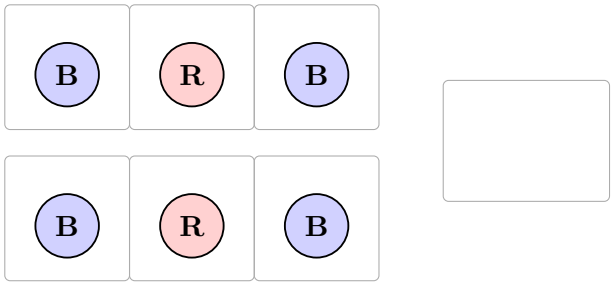
Each instruction names a slot and a color. Set that slot to that color on both boards, even if it is already that color. Slots are 1, 2, 3 from left to right.

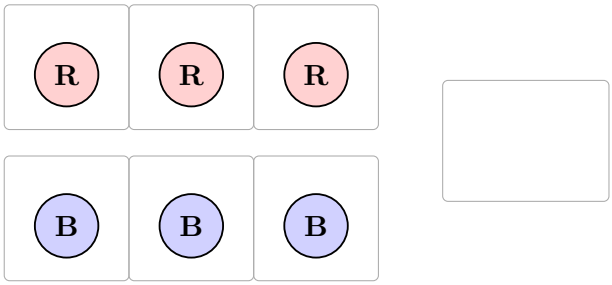


Problem 1: How does the fewest number of instructions needed for a match depend on the starting pair? Develop a rule that works for every pair.

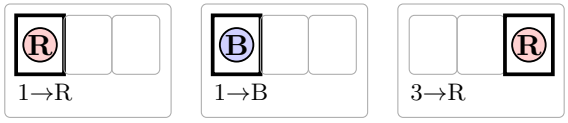


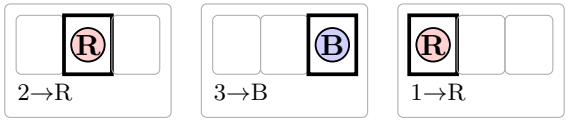


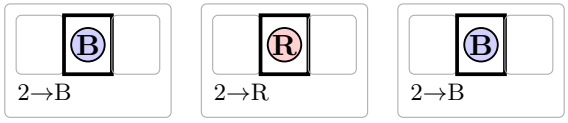




Problem 2: For each instruction story, how many different final boards can come from the eight possible starts? Determine exactly which starting colors still matter.







Problem 3: Explain exactly when an instruction story forgets every starting color. Could two different starts agree before that happens?